



“Over the last year-and-a-half, we’ve seen rapid gains in quantum computing in terms of the technology stack and algorithmic advances in multiple application areas”

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“Investor interest is rising sharply, with Indian quantum start-ups raising more than \$40 million between January and October 2025”

DEVROOP DHAR
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50-100 physical qubits in five years, and 50-1,000 physical qubits in eight years across platforms such as superconducting and photonic technologies,” says Jagdish Bhandarkar, Chief Disruption Officer at Deloitte South Asia. But building practical quantum systems requires specialised cryogenic infrastructure (engineered networks designed to store, transfer, and manage liquefied gases at ultra-low temperatures), advanced fabrication capabilities, precision control electronics, and deep semiconductor supply chains, areas where India still trails global leaders.

India has nevertheless started showing early signs of hardware progress, including work on a 6-qubit processor and private-sector milestones such as QpiAI’s 25-qubit system. “These are important steps, but global leaders are already operating at much larger qubit counts and focused on logical qubits, error correction, and scalable

GLOBAL EFFORTS

Committed Public Investment



China

\$15 billion (National Quantum Initiative) + RMB 1 Trillion (National Venture Guidance Fund)



United States

\$1.8–2.5 billion (National Quantum Initiative/NQI Reauthorization) + \$7.1 Billion (Federal Post-Quantum Cryptography/PQC Migration)



EU

€1 billion (Quantum Flagship) + Synergy with €43 Billion European Chips Act



UK

£2.5 billion (National Quantum Strategy)



Japan

JPY 130 Billion quantum-specific (nested within JPY 1.05 Trillion broad R&D budget) + JPY 50 Billion quantum startup industrialisation push



India

₹6,003 cr National Quantum Mission

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architectures,” adds Bhandarkar. “Commercially viable, fault-tolerant quantum machines are still not available at scale globally. So, India is not late, but it is early in the race. The next five years are crucial.”

The economics are equally challenging. A mid-range 20-100 qubit system could cost between \$1.5 million and \$15 million, while flagship enterprise-scale systems may cost \$50 million to \$100 million.

BUILDING ON STRENGTHS

Analysts say India may not necessarily need to replicate the US or China playbook. According to Singhee of IBM the quantum stack includes hardware, software, algorithms, and applications. India has strong potential to lead in areas such as quantum algorithms, software, and application discovery, leveraging its strengths in mathematics, software engineering, and scientific talent.